

Department: Data Science and Artificial Intelligence
Course Name: Data Science Tools and Software
Course Code: 03203N
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Level: Two



Sheet 4
(Descriptive Statistics)

Solve the following questions using Python :

- (1) Read data from a CSV file named "cars_data.csv" into a Pandas DataFrame?**

- (2) Rename the column "Unnamed: 0" to "car model"?**

- (3) Check the structure of a Pandas DataFrame, including the number of entries, column names, and data types?**

- (4) Calculate the mean of all numerical columns in a DataFrame. Additionally, how can you compute the mean for each row instead of each column?**

- (5) Compute the sum of all numerical values in each column. Additionally, compute the sum for each row instead of each column?**

- (6) Find the median and mode of all numerical columns.**
- (7) Determine the minimum and maximum values for each numerical column. Additionally, how can you identify the car model with the maximum value in any categorical column?**
- (8) Find the index of the row that contains the highest value in the "mpg" (miles per gallon) column.**
- (9) Determine the 25th percentile, 50th percentile, and 75th percentile of the "mpg" (miles per gallon). Also, how can you confirm that the median value matches the 50th percentile?**
- (10) Determine the frequency of unique values in the "am" column to understand how many times each value appears.**
- (11) Obtain a statistical summary of all numerical columns in a DataFrame, including measures like count, mean, standard deviation, minimum, and quartiles.**
- (12) Generate a summary of the "mpg" column, including its count, mean, standard deviation, and quartiles, in a single command.**

- (13) How can you retrieve the five-number summary (minimum, first quartile, median, third quartile, and maximum) for the "mpg" column in a DataFrame?
- (14) Compute the mean value of the "mpg" column.
- (15) Find the median values of both the "mpg" and "hp" columns together.
- (16) Generate a summary of key statistics (such as count, mean, standard deviation, min, and quartiles) for only the "mpg" and "hp" columns.
- (17) Compute the range of the "mpg" column.
- (18) Compute the variance of the "mpg" column.
- (19) Compute the standard deviation of the "mpg" column.
- (20) Compute the coefficient of variation (CV) for the "mpg" column

Good Luck
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